

Yang Zhang

Quantitative Finance | AI/LLM Engineering | Machine Learning | Data Analytics
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EDUCATION

University of Michigan <i>Master of Science in Quantitative Finance and Risk Management</i> Relevant Coursework: Machine Learning for Finance I/II; Advanced Financial Mathematics I/II; Stochastic Processes; Stochastic Analysis; Statistical Learning; Financial Data Analysis; Computational Finance; Algorithmic Trading	Ann Arbor, Michigan, USA	Sep 2026 – Jun 2028 (Expected)
Shanghai University <i>Bachelor of Engineering in Data Science and Big Data Technology</i> Relevant Coursework: Machine Learning; Neural Networks & Deep Learning; Natural Language Processing; Computer Vision; Data Mining; Operations Research & Optimization; Information Theory & Coding; Big Data & Cloud Computing	Shanghai, China	Sep 2022 – Jun 2026 (Conferred) GPA: 3.57/4.00

PROFESSIONAL EXPERIENCE

Ernst & Young (EY) <i>AI Agent / Full-Stack Developer Intern</i> <i>Developed two AI systems: a multi-agent news intelligence platform and an Outlook assistant for email and meeting workflows.</i> - Designed a ReAct-based parallel multi-agent pipeline whose topic-specific agents used OpenAI function calling to select RSS/web-search tools, collect articles concurrently, score and verify URLs, and stream execution traces through FastAPI/SSE. - Implemented hybrid RAG with OpenAI embeddings, BM25/vector search, semantic deduplication, and ChromaDB; persisted articles, RAG traces, agent events, and deduplication decisions with SQLite/SQLAlchemy, and packaged reusable MCP tools with Docker. - Developed an Azure-hosted Outlook Add-in backend using OpenAI and Microsoft Graph for email summaries, reply drafts, meeting extraction, availability checks, and Teams creation; secured delegated access with Entra ID and OAuth 2.0/OIDC.	Shanghai, China	Mar 2026 – Jun 2026
China UnionPay Data Co., Ltd. <i>Data Analyst Intern</i> <i>Automated credit-card reporting and bank benchmarking, converting monthly operating data into decision-ready dashboards.</i> - Automated operating reports from SQL-derived BI exports and Excel using Python, Pandas, NumPy, and Matplotlib; built Power BI dashboards for active cards, balances, utilization, customer segments, installment products, and campaign performance. - Analyzed monthly and annual performance across 9 partner banks and 16 listed banks, building reusable benchmarks for account-age trends, customer value, credit limits, credit quality, transaction volumes, and complaint ratios.	Shanghai, China	Jul 2024 – Aug 2024

PROJECT & RESEARCH EXPERIENCE

Weakly Supervised Video Anomaly Detection <i>Computer Vision / Deep Learning Engineer</i> <i>Built a weakly supervised system that localizes surveillance-video anomalies from video-level labels and supports web inference.</i> - Benchmarked C3D (4096-D) and Kinetics-pretrained 3D-ResNet-101 (2048-D) features across 32 temporal segments per UCF-Crime video, obtaining baseline frame-level AUCs of 0.6927 and 0.7854 for subsequent model comparison. - Developed PyTorch detectors combining Dense BCE, MIL ranking, magnitude-contrastive loss, top-$k = 5$ selection, multi-scale dilated temporal Conv1d, LayerNorm, dropout, and temporal-attention experiments. - Improved AUC to 0.8366 for the best single model and 0.8479 for a four-model ensemble; built Python/OpenCV web inference for uploaded videos or feature files, mapping 32 segment scores back to frame and time ranges.	Senior Thesis, 2026
AI-Powered Course Assistant System <i>Full-Stack LLM Developer</i> <i>Built an AI teaching platform that answers from course materials, retrieves related knowledge, and responds through a digital human.</i> - Developed a grounded course-material Q&A backend using Ollama, DeepSeek-R1, and BGE-Large-ZH-v1.5 embeddings. - Implemented LightRAG with entity/relation extraction, low- and high-level retrieval, and incremental knowledge-base updates; visualized retrieval graphs with NetworkX/Pyvis/Streamlit and integrated APIs with a Vue 3 and Axios interface. - Integrated a Three.js digital human with Blender assets, AnimationMixer, and Azure TTS; resolved model-loading and animation-timing issues while maintaining 30 FPS rendering and sub-3-second responses in project tests.	Feb 2025 – May 2025
MindSpore Innovation Camp <i>NLP / LLM Engineer</i> <i>Applied modern NLP and LLM methods in model training, evaluation, and local chatbot deployment, earning Second Prize.</i> - Trained and evaluated Transformer and BERT models with MindSpore/MindNLP, Huawei Cloud ModelArts, and an NVIDIA P100 for sequence modeling, sentiment classification, inference, and BLEU evaluation. - Deployed a local ChatGLM-6B chatbot with Gradio; explored PEFT, prompt tuning, instruction tuning, and deployment workflows.	Mar 2024 – Apr 2024

PUBLICATION

Customer Segmentation: Data-Driven Research on Transaction-Level Data of JD.com <i>First Author</i> <i>Led a first-author study using JD.com behavioral and transaction data to segment customers and model product preferences.</i> Engineered customer, product, price, membership, and click-behavior features with Python/Pandas/NumPy; compared K-means, Logistic Regression, Decision Tree, Random Forest, and Stacking using RFE and hyperparameter search.	ICCSMT 2024, EI Indexed
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SKILLS

Programming / Data: Python, C/C++, SQL, JavaScript, HTML/CSS, MATLAB, Pandas, NumPy, Matplotlib, Power BI
AI / LLM / ML: PyTorch, MindSpore, BERT, Transformer, AI agents, tool calling, RAG, embeddings, BM25/vector/hybrid retrieval, LightRAG, ChromaDB
Backend / Cloud: FastAPI, REST/SSE, Azure, Microsoft Graph, Entra ID, OAuth 2.0/OIDC, Docker, SQLAlchemy, SQLite/MySQL, Git